

Joint Structure and Function

Joint Design -

- A person with a lower extremity amputation can order limbs made of different materials, depending on their functional demands.
- Springlike materials, which are harder to stretch & which recoil more, are used for activities with large loads and high speeds (Ex - running, jumping).
- The materials with less elastic recoil are used for walking.
- A track athlete with lower extremity amputation can order feet that are less stable but have greater recoil.
- The composition of human joints is determined by a number of factors, including genetic expression, cellular interaction during development & functional demands.

Basic Principles -

- A joint connects 2 or more components of a structure.
- The design of a joint and the materials used in its construction depend partly on the function of the joint & partly on the nature of the components.
- Joints that provide stability have a different design than those that provide mobility.

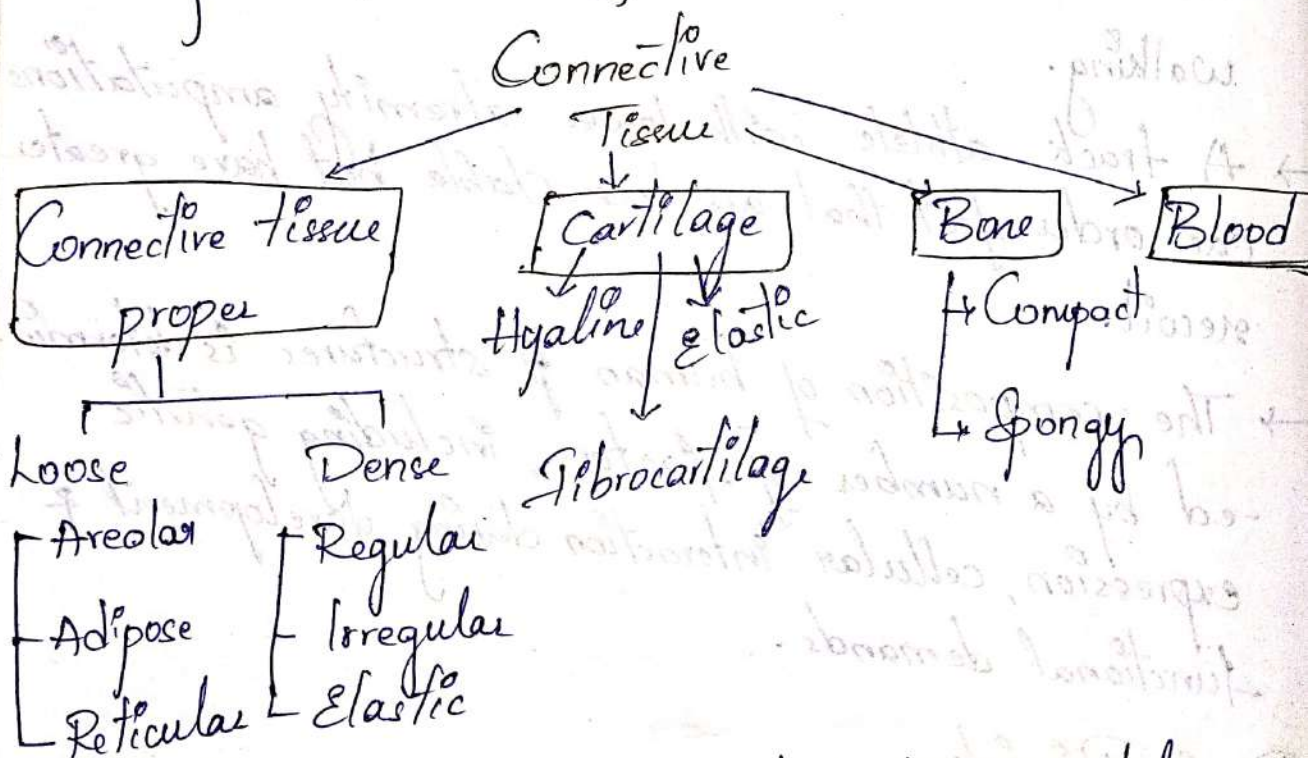
→ The complexity of the design & Composition matches the range of functional demands - The more varied the demands, the more complex the design.

Materials Found in Human Joints

Structure of Connective tissue -

→ It structures include many connective tissues - bones, bursae, Capsules, Cartilage, disc, fat pads, labra, menisci, plates, ligaments and tendons.

→ They are 4 classes of Connective tissue:-



→ Connective tissue is characterized by widely dispersed cells and a large volume of Extracellular matrix.

→ Extracellular matrix has both
Interfibrillar Component
Fibrillar Component

→ Nerves and muscles depends on cell structure & function.

Cells -

→ Cells of all connective tissue derive from Mesenchymal precursor cells.

→ Two types of Connective tissue cell.

1) Fixed 2) Transient

→ ① Fixed -

→ Fibroblast -

- It is a basic cell of most connective tissue

- Produces Extracellular Matrix.

- Depends on mechanical & physiological environment

Location - Found in tendon, ligament, skin, bone etc

Function - Creates mostly type I Collagen

→ Chondroblast (cartilage) -

Location - Found in cartilage.

Function - Produces mostly type II Collagen.

→ Osteoblast (bone) -

Location - Found in bone.

Function - Produces type I Collagen & hydroxyapatite.

→ Osteoclast -

Location - Myt Monoclu - derived, found in bone.

Function - Responsible for bone resorption.

→ Mast Cells -

Location - Found in various connective tissues

Function - Inflammatory mediators.

→ Adipose cells -

Location - Found in adipose tissue

Function - Produce & store fat.

→ Mesenchyme cells -

Location - Found primarily in Embryos & bone marrow

Function - Any Connective tissue cell can be differentiated.

② Transient

→ Lymphocytes -

White blood cells that have surface proteins specific for antigens.

→ Neutrophils -

WBC involved in fighting infection.

→ Macrophages -

Derived from monocytes, move into specific tissue, involved in immune response.

→ Plasma Cells -

B lymphocytes producing antibodies.

Extracellular Matrix -

→ It is the part of connective tissue outside the cells.

→ It contains proteins and water.

Fibrillar Component -

→ It contains of 2 major classes of structural proteins.

i) Collagen

ii) Elastin

Collagen -

→ It is the main substance of most connective tissues, found in all multicellular organisms.

→ Most abundant protein in the human body is of 25% to 30%.

Collagen type -

Classification	Type	Common locations
Fibrillar	I	Tendon, bone, joint capsule, skin, ligament, fibrocartilage, meniscus. Accounts for 90% of body Collagen.

II

Cartilage, intervertebral disc, Hyaline cartilage.

III

Joint capsule, skin, blood vessels, tendons, ligament.

V

Cartilage, tendon

XI

Cartilage, other tissue (found with type V).

IX

Cartilage, cornea (found with type II).

Classification	Type	Common location
Fibril-associated	<u>XII</u>	Tendon, Ligament (found with type I)
	<u>XIV</u>	Fetal skin & tendon
	<u>IV</u>	Basement membrane
Network formation	<u>X</u>	Hypotrophic Cartilage
	<u>VIII</u>	Unknown
Filamentous	<u>VI</u>	Blood vessels, skin
Anchoring	<u>VII</u>	Anchoring filaments

→ Basic building block of collagen is a triple helix of 3 polypeptide chains called Tropocollagen molecule.

→ Collagen peptide chains are synthesized in the rough endoplasmic reticulum inside the fibroblasts & then move through the cell towards the cell membrane.

→ Tropocollagen molecules aggregate to form microfibrils in turn combine to form fibrils (fibers) of varying size.

→ Intermolecular and intermolecular cross-links stabilize and strengthen the enlarging fibrils (fibers).

→ Fibrils enlarge as more collagen is added to them and more cross-linking.

→ Older fibrils are larger and contain more cross-links, making them stronger.

→ Relaxed tissue, the fibers have a wavy configuration called a Crimp.

→ As the collagen fibers stretched, the Crimp disappears.

Elastin -

→ Found in many connective tissue and the molecule consist of single Alpha like strands without a triple

helix.

→ Alpha-like strands are cross-linked to each other to form rubber like, elastic fibers.

→ Elastic fibers branch freely. And found in all pts structures as well as in skin, tracheobronchial trees and walls of arteries.

→ Elastin makes up a much smaller portion of the fibrous component in the extracellular matrix than Collagen.

→ Aorta - 30% Elastin
20% Collagen

→ Ligamentum Nuchae - 75% Elastin
15% Collagen

→ Achilles tendon - 4.4% Elastin
86% Collagen

Interfibrillar Component

→ It contains water and proteins, primarily glycoproteins and proteoglycans (PGs).

→ Glycoprotein is a protein with a carbohydrate attached.

→ There are thousands of glycoproteins in the body.

→ The term "Glycoprotein" includes PGs.

→ Glycoprotein found in all tissues.

→ PGs found in Connective tissues.

→ PGs are also known as Mucopolysaccharides.

→ Interfibrillar matrix referred as Ground Substance.

→ Ground Substance is a mixture of PGs and water.

Class Proteoglycans -

Classification - Large Extracellular aggregation

Name

- Versican

Location/Composition/

- Found in smooth muscle cells,

Function

- fibroblasts.

Classification	Name
Large Extracellular Aggregating	Versican Aggrecan Brevican Neurocan Decan Biglycan
Small leucine-rich proteoglycans (SLRPs)	Decorin Biglycan

Location, Composition and Function.

Found in smooth muscle cells, fibroblasts.

Found in numerous chains of KS & CS (Chondroitin sulfate Keratan sulfate)

Binds to hyaluronan.

Creates osmotic swelling pressure in cartilage by attracting water.

Found in Nervous system; cell adhesion & migration.

Found in Nervous system; cell adhesion & migration.

1 or 2 Chondroitin Sulfate or Dermatan Sulfate chains.

Binds & regulates growth factors, modulates cell functions, regulates collagen fibrillogenesis, interacts with collagen types I, II, III, V, VI, XII, XIV.

2 GAG chains containing Chondroitin Sulfate or Dermatan Sulfate.

Directs type VI collagen network assembly.

Binds to complement & transforming growth factor beta (TGF-β).

Fibromodulin

Epiphyscan

Lumican

Serglycin

Syndecans

Betaglycan

CD44 family

Thrombospondin

1 Keratan Sulfate GAG chain.

Interacts with type I & II Collagen, binds to growth factors.

Found in epiphyseal cartilage.

Similar to fibromodulin, found in cornea, muscle, intestine, cartilage.

Protein core of heparin: PGs regulate enzyme activities

Secretory granules.

Cell-transmembrane PG containing HS acts as a receptor

for heparin-binding factors.

Contains Heparan Sulfate & Chondroitin Sulfate

Binds TGF- β .

Cell surface receptor for hyaluronan

Binds to thrombin

Basement
membrane
PGs

Perlecan
HS + CS PGs
Found in all tissues; function uncertain

Nervous tissue
PGs

Bamacan
Phosphacan
Aggri
NG2 PG
Sound in various tissues;

Aggri
PGs

Nervous tissue cell adhesion
Aggregates acetylcholine receptors
Sound in developing cells.

Note: CS
KS
DS
GAG
HS
PG

CS - Chondroitin Sulfate
KS - Keratan Sulfate
DS - Dermatan Sulfate
GAG - Glycosaminoglycan
HS - Heparan Sulphate
PG - Proteoglycan.

CS
KS
DS
GAG
HS
PG

Chondroitin Sulfate
Keratan Sulfate
Dermatan Sulfate
Glycosaminoglycan
Heparan Sulphate
Proteoglycan (GAG)

Glycoaminoglycans (GAGs)

GAG	Localization	Comments	Composition
Hyaluronan	Synovial fluid, Vitreous humor, loose CT, healing CT, Cartilage	Forms large PG aggregates	Glucuronic ^o uronic acid Glucosamine
Chondroitin Sulfate	Cartilage, bone, heart valves, tendons, ligaments	Most abundant GAG, red with compression	Glucuronate Glucosamine with 4-Sulfate / 6-Sulfate
Heparan Sulfate	Basement membranes, cell surfaces	Interacts with numerous proteins	Glucuronate Glucosamine Variable Sulfation
Heparin	Intercellular granules in mast cells lining arteries	Key structural unit is 3-glucosamine + 2-glucuronate	Glucuronate, iduronate Glucosamine Variable Sulfation
Dermatan Sulfate	Skin, blood vessels, tendons, ligament	red with tensile stress	Iduronate Galactosamine

Galactose
Glucosamine
Forms part of large PG
aggregates in cartilage.

Cornea, bone, Cartilage
CT - Connective tissue
PG - Proteoglycan

Keratan
Sulfate

Note :-

- Carbohydrate portion of PG consists of long chains of repeating disaccharide units called as Glycosaminoglycans (GAGs).
- The major types of sulfated GAGs are Chondroitin + and Chondroitin 6 Sulfate, Keratan Sulfate, heparin, heparan Sulfate and dermatan Sulfate.
- 1 large cartilage PG is called Aggrecan.

→ Glycoproteins such as fibronectin, laminin, chondronectin, osteonectin, tenascin, tenomodulin, chondromodulin and entactin play an important role in components of the extracellular matrix.

* Proteoglycan Characteristics :-

→ PG are distinguished by their protein core & by their attached GAGs.

→ PG attract water through their attached GAGs.

→ PG regulate collagen fibril size.

→ PG may attach to hyaluronate (another GAG) to form large aggregating structure.

→ And are increased in tissues subjected to alternating cycles of compression.

Extracellular Matrix Structure & Function :-

→ Extracellular matrix of connective tissue determines its function.

→ The type & proportions of components create the different tissues.

* Interfibrillar Component : PG (protein + GAG), glycoproteins.

* Fibrillar Component: Collagen (mainly Type I & II), Elastin.

Specific Connective Tissue Structures:

Ligaments

→ Ligaments connect one bone to another, near a joint.
→ Ligament may blend with the jt capsules & appear as thickenings in the capsule (e.g. - anterior band of the inferior glenohumeral ligament).

→ Other ligaments are distinct, often appear as distinct dense white bands or cords of connective tissue.
Ex - Anterior Cruciate Ligament.

→ Ligaments are heterogeneous structures.
- a small amount of cells (10 to 20% of tissue volume, mainly fibroblast).

- Large amount of Extracellular matrix
(80% to 90% of tissue volume).

→ PGs constitute about 0.2% of tissue dry wt & contain primarily dermatan sulfate GAGs.

→ Fibrillar Components of the extracellular matrix, most ligament is mainly composed of type I collagen and lesser amount of type III, IV, V Collagen.

→ Type I Collagen fibrils in ligaments are densely packed, with fiber bundles arranged in line.

- Arrangement of Collagen fibers & the Collagen or elastin fiber ratio, In various ligament to provide stability & mobility for a particular jt.
- Collagen fibres in ligaments run in many directions enabling the ligament to resist forces in more than one direction.
- Tendons & Ligaments may insert directly into bone via fibrocartilage or indirectly via fibrous attachments.

Tendons

- Tendons ~~cannot~~ connect muscle to bone & transmit forces developed by the muscles to their bony attachments.
- Each muscle has tendinous material interspersed b/w it and bone.
- Tendons contain mostly type I Collagen (more 95%) with type II in the tendon sheaths.
- The type I Collagen in tendons is thought to be an adaptation to large tensile forces.
- Tendons contain water, PGs, that contain mostly dermatan sulfate, & other glycoproteins.
- Endotendon, containing mostly Type III Collagen, fibrils also encloses the nerves, lymphatic vessels, & blood vessels.
- Sheath enclosing the entire tendon is called the epitenon.

→ Paratenon is loosely attached to the outer surface of the epitenon.

→ The Epitenon & Paratenon together called the peritendon.

→ Paratenon protects the tendon from friction.

→ Tendons have 2 types of bone attachments
fibrocartilaginous & fibrous.

Tendon Structure -

First Zone : Consists of tendon proper & similar to the tendon midsubstance.

Second Zone : Consists fibrocartilage

Third Zone : Consists mineralized fibrocartilage

Fourth Zone : Consists of bone.

Bursae

→ Bursae are flat sacs of synovial membrane in which the inner sides of the sacs are separated by fluid film.

→ They are found where moving structures are in tight approximation i.e., tendon & bone, bone & skin, muscle & bone or ligament & bone.

Subcutaneous bursae : Found b/w patella & skin and b/w olecranon process of ulna & skin.

Subtendinous bursae : Lie b/w tendon & bone.

Submuscular bursae : Lie b/w muscle & bone.

Fibrocartilage

Bone

→ Bone is the hardest of all the connective tissue.

→ The organic material gives bone its flexibility & tensile strength.

→ Inorganic material gives bone its compressive strength.

→ Bone cells includes : Fibroblasts

Osteoblasts

Osteocytes

Osteoclasts

progenitor cells. (differentiate into osteoblasts).

→ Osteoblasts are the primary bone-forming cells that are responsible not only for synthesis of bone but also for its deposition & mineralization.

→ Osteoclasts are responsible for bone resorption.

→ Bone has 2 layers dense outer layer called Compact or cortical bone, inner layer is called Cancellous bone.

→ Periosteum is a fibrous layer that covers the entire surface of bone except articular surface.

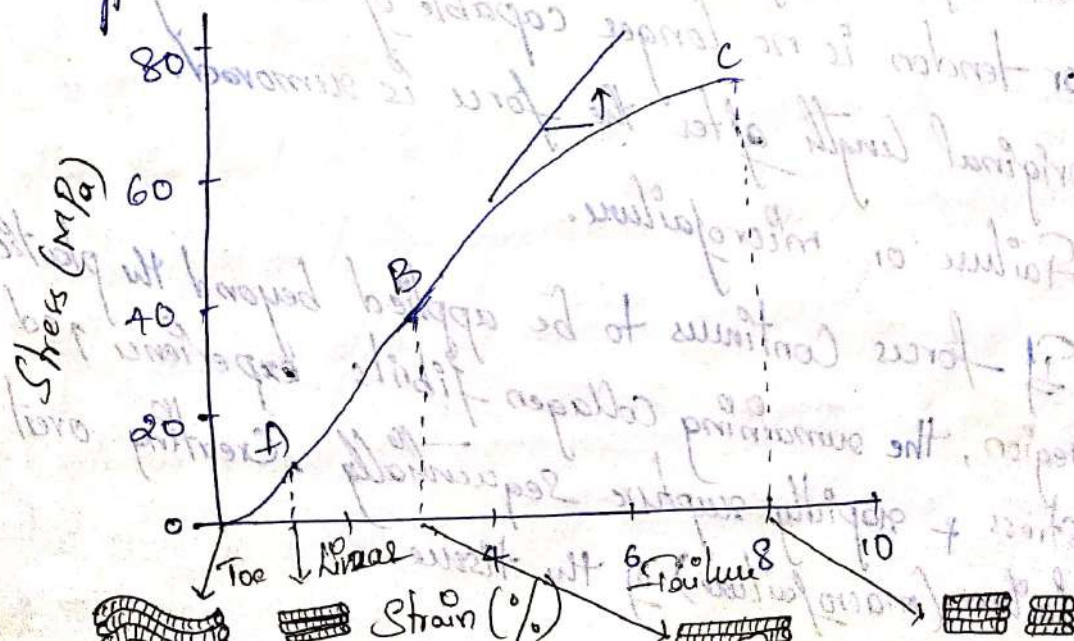
- Periosteum acts as a reservoir for cells that are needed for growth & repair.
- Bone serves physiological as well as mechanical functions.
- Imbalance b/w bone synthesis & resorption in which the osteoclasts break down or absorb the bone faster than osteoblasts can remodel the bone results in condition called Osteoporosis.
- Bone mineralization may also be decreased while cells continue to synthesize other elements this is called Osteopenia.

General Properties of Connective Tissue :-

Mechanical Behavior :-

Load Deformation and Stress - Strain Curves

- Each material has its own unique stress-strain curve.
- A typical curve for tendon and ligament.



→ 1st region of Curve (O to A) is toe region.

→ In this region, a minimal amount of force produces a relatively large amount of deformation.

→ Stress is low

Strain is typically in 1% to 2% range.

→ 2nd portion of Curve (A to B) is elastic region.

→ In which Strain (elongation) has a linear relationship with stress.

→ In this region of Curve, Collagen fibrils are being stretched & are resisting the applied force.

→ When the load is removed, the ligament or tendon will return to its prestressed dimensions, but it takes some time.

→ Stresses and strains that occur with normal activities & typically extends to about 4% strain.

→ 3rd Portion (B to C) is plastic region.

→ The failure of collagen fibers begins, & the ligament or tendon is no longer capable of returning to its original length after the force is removed.

→ Failure or microfailure.

→ If forces continue to be applied beyond the plastic region, the remaining collagen fibrils experience increased stress & rapidly rupture sequentially, creating overt failure (macrofailure) of the tissue.

- Ligament or tendon failure occurs in the middle of the structure through a disruption of the connective tissue fibers called a Rupture.
- Failure occurs at a bony component / attachment of the ligament or tendon is called a avulsion.
- Failure occurs within bony tissue, called fracture.

Time - Dependent and Rate - Dependent Properties.

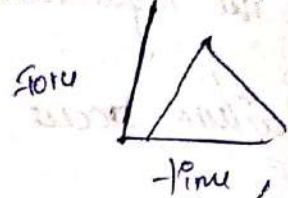
- Viscous qualities make the deformation & return time - dependent.
- Viscoelastic material possesses characteristics of Creep, Stress - relaxation, Strain rate sensitivity, Hysteresis.

Creep

- Force is applied to a tissue and maintained at the same level while the deformation produced by this force is measured, the deformation will gradually ↑.
- Force remains constant while length changes.
- Connective tissues also will gradually creep (elongate) after the load & initial elastic response to a constant tensile load and then gradually return to their original length (recovery) after the load is removed.
- Cartilage & bone, and Compressive loading is used to test creep & to the depth of indentation represents creep & recovery.

Stress - Relaxation

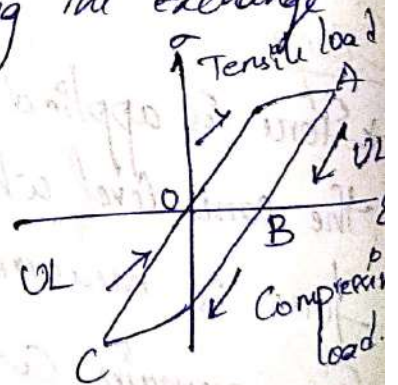
- If a tissue is stretched to a fixed length while the force required to maintain this length is measured, the force needed will $\downarrow$ over time.
- Length remains constant while force $\downarrow$.



Hysteresis

- Force & length of the tissue are measured as force is applied (loaded) & removed (unloaded)
- Resulting load-deformation curve do not follow the same path.

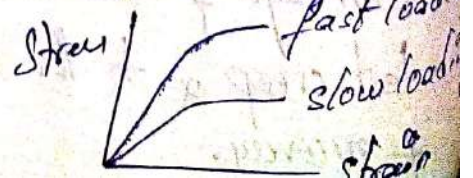
- Some energy loss usually as heat.
- Energy gained as a result of the lengthening work (force $\times$ distance) is recovered during the exchange from energy to shortening work.



Strain - Rate Sensitivity

- Load is applied rapidly, the tissue is stiffer & a larger peak force can be applied to the tissue than load was applied slowly.
- Stress relaxation also will be larger than if the load was applied slowly.

- Creep take longer to occur under condition of rapid loading.



Joint Function :-

- Synarthrodial jts are simple in design & function primarily as stability jt.
- Diarthrodial jts primarily provide mobility & also stability.

Kinematic Chain

- 2 types of Kinematic chain
 - Open Kinematic chain
 - Closed Kinematic chain

Open Kinematic Chain :- distal end of the chain is free to move & 1 jt can move independently of others in the chain.

Closed Kinematic Chain :- Both the proximal & distal ends of the chain remain fixed.

Joint Motion :-

Range of Motion

- The normal ROM of a jt is sometimes called the anatomic or physiologic ROM.

- Anatomic range is determined by a number of factors:

Shape of jt surface

Jt capsule

Ligaments
Muscle bulk
Musculotendinous
Bony Structures.

- ROM is limited by soft tissue structures.
- Sensation experienced by the examiner performing passive physiologic movements at each jt is referred to as the End-feel.
- ROM exceeds the normal limits, the jt is hypermobile.
- ROM is less than what the normally be permitted by the structure, the jt is hypomobile.
- Hypermobility is caused by a failure to limit motion through bony elements or soft tissues, including weak musculature, leading to instability.
- Hypomobility is caused by bony or cartilaginous blocks to motion or by inability of the capsule, ligaments or muscles.
- Contracture, shortening of soft tissue structures around a jt, is one cause of hypomobile.

Osteokinematics:

- It is a rotary movement of the bones in space during physiological jt motion.
- Movements of the bony axes in Sagittal, frontal,

transverse planes that occurs at jt.

Arthrokinematics:

→ Physiological jt motion involves rotation of bony segments as well as motion of the jt surface in relation to another.

→ Arthrokinematics, is used to refer to these movements of jt surfaces relative to one another.

→ Roll, the rolling of 1 jt surface on another, as in a tire rolling on the road.

→ Ex :- Knee,

Femoral condyles roll on the fixed tibial surface during the initial stages of knee flexion & extension in standing.

→ Sliding, Pure translatory motion, gliding over one component over another. (Braked wheel skids.)

→ Spin, Ex :- Proximal phalanx slides over the fixed end of the metacarpal during flexion & extension.

→ Spin, Pure rotatory motion, Rotation of the movable components, as well as top spins.

→ Ex :- Elbow,

Head of the radius spins on the capitulum of the humerus during supination and pronation of forearm.

→ Rolling & Sliding usually occur in order to maintain jt integrity.

Convex-Concave Rule:

→ Concave articulating surface is moving on a stable convex surface, sliding typically in same direction.

Convex jt surface moves on a concave surface, the bony typically rolls in one direction, & glides in another direction / opposite direction.

→ It fits into either an Ovoid & Sellar category.

→ Ovoid jt, one surface is convex & the other surface is concave.

→ Sellar jt, each jt surface is both convex & concave.

→ If the articular end of the bone is not free to move of the bone moves in the appropriate direction, then when the distal end of the bone moves, abnormal force will be created in the jt.

→ Articular surfaces to be free to move in the appropriate direction as the bony lever rotates, the jt must have a certain amount of jt play.

→ This freedom of movement of 1 articular surface on another can be tested by an examiner when it is in a loose-packed position.

- If structure are lax (loose), the jt may have too much play & become unstable.
- If structure is tight, the jt may have too little play movement.
- The axis of rotation at any particular point in the jt motion is called the instantaneous axis of rotation (IAR).
- All synovial jts have a close-packed position in which jt surfaces are maximally congruent & the ligaments & capsule are maximally taut (tight).
- Close-packed position is at the extreme end of ROM.
- No jt play is possible. In humeroulnar, knee and interphalangeal jt.
- Loose packed position of jt, articular surface are relatively free to move in relation to one another.
- In loose packed position, the jt has a maximum amount of jt play.

Biomechanics

① Viscoelasticity :-

→ Elasticity refers to the materials ability to return to its, original state after deformation, after removal of deforming load.

→ The elastic qualities, in connective tissues depends on Collagen and elastin content and organization.

→ The elasticity implies that length changes
a. deformations are directly proportional to applied forces (or) loads and do not exhibit time dependent behaviour.

→ Viscosity refers to a materials resistance to flow which depends on PG and water composition of tissue.
↳ Proteoglycans

→ It exhibits time and state dependent properties.

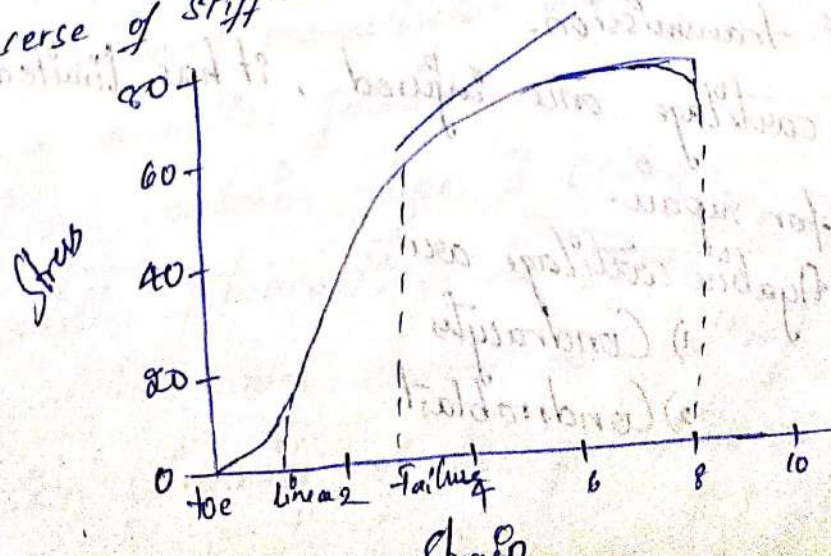
→ Viscosity increases as pressure increases and decreases as temperature rises.

② Young's Modulus :-

→ Mechanical behaviour of the material & it is a measure of mechanical stiffness.

Measure of mechanical stiffness and resistance is Compliance.

→ The inverse of stiffness and resistance is Compliance.



(2) Stress :- * force within the material are produced to oppose the applied force.
unit of pascals or Megapascals
($P = N/m^2$) ($MPa = N/mm^2$)

→ Used to measure - pressure
Force per unit area.

→ When load is applied to any structure or material, any internal resistance to that deformation is produced.

$$S = F/A$$

S - Stress
F - Applied force
A - Area.

Strain :-

→ Percentage change in the length or cross-section of a structure or material.

→ Calculated by mathematically formula

$$S = (L_2 - L_1) \div L_1$$

L_1 - Original length
 L_2 - final length.

(4) Hyaline Cartilage :- (Articular)

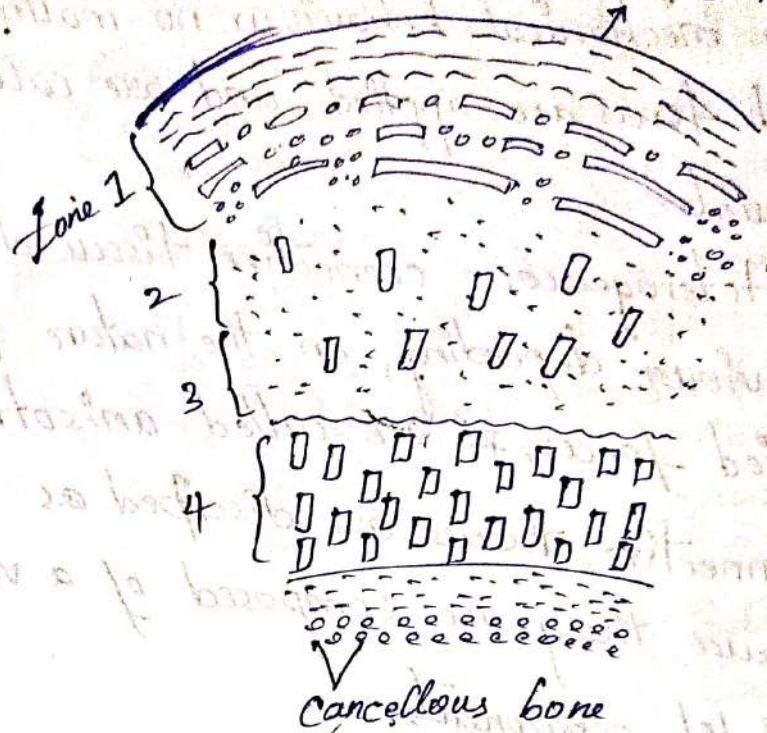
→ It forms the covering of end of bones for most of the joints. (Synovial joints)

→ It has smooth and low friction surface for the articulation of one bone to another bone and also for weight bearing and weight transmission.

→ Hyaline cartilage once injured, it has limited and improper mechanisms for repair.

→ Cells in Hyaline cartilage are
1) Chondrocytes
2) Chondroblast

- Chondrocytes produce - GAGs, PGs, collagen and link proteins
- Chondroblast produced in active cells absorbed in non-growing tissues.



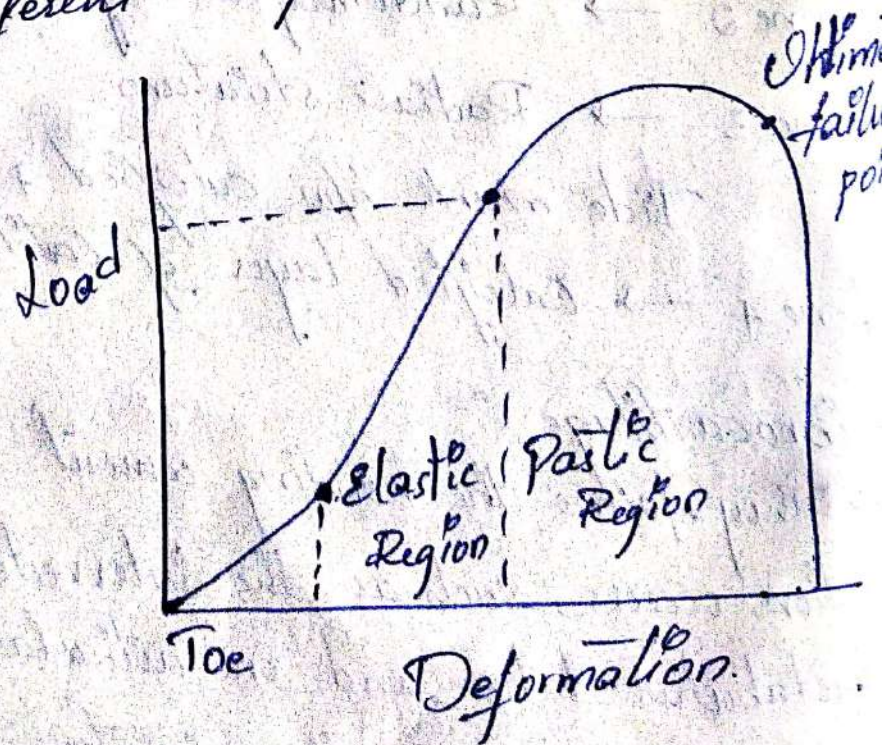
- Zone 1 → Reduce friction & arranged in parallel
- Zone 2 → Randomly arranged collagen fibers
- Zone 3 → Radiate striations
- Tide mark b/w calcified & uncalcified cartilage.
- Zone 4 → Calcified layer of cartilage.

Fibrocartilage :-

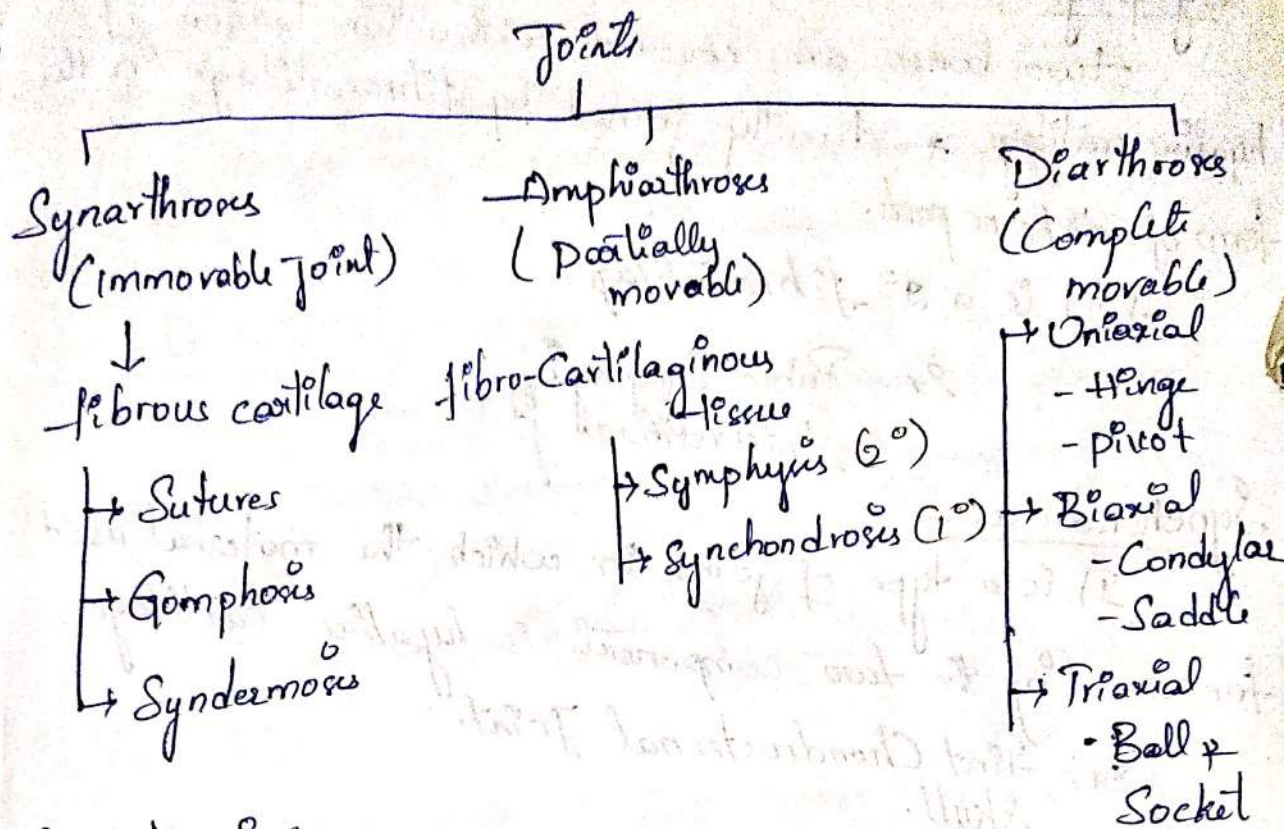
- They form the bonding cement in joints where little motion occurs, such as, the intervertebral discs, the glenoid & acetabular labra and one articular surface of the temporomandibular & sacroiliac joints.
- fibrocartilage contains type I collagen.
- Yellow elastic fibrocartilage is found in the ears & epiglottis and has more elastin than white fibrocartilage which consists primarily of collagen fibers.

General properties of Connective tissue.

- Homogeneous materials, such as steel, display the same mechanical behaviour no matter the direction in which forces are applied and are called as isotropic materials.
 - Heterogeneous connective tissue display very different behaviours, depending on the nature & direction of applied forces, & are called anisotropic.
 - Connective tissues are described as heterogeneous because they are composed of a variety of solid & semisolid components.
 - Function of a structure depends on a combination of the properties in the different components.
- Mechanical behaviour:



Classification of joint :-



Synarthrosis :-

Sutures -

Coronal Sutures :- The serrated edges of the parietal & frontal bones of the skull are connected by a thin fibrous membrane to form the coronal suture.
Ex - Skull

Gomphosis -

Connected by fibrous tissue.

It is a peg and socket junction b/w the tooth & its socket. Periodontal ligament connect the dental element to the alveolar bone.

Ex - tooth



periodontal ligament.

Syndesmosis -

It is a type of fibrous joint, where two bones are connected with an interosseous ligament. It forms a little movement.

Ex - Radio ulnar joint
Tibia & fibula joint

Amphiarthrosis:-

Symphysis -

- Two bones are covered with a thin lamina of hyaline cartilage & directly joined by fibrocartilage in the form of disks or pads.

→ It is a 2° fibrocartilage

Ex - Pubic Symphysis
Intervertebral jt.

Synchondrosis -

It is a type of joint in which the material used for connecting the two components is hyaline cartilage.

Ex - First Chondrosternal Joint.
Skull.

Diarthrosis:-

Synovial joint, the ends of the bony components are free to move in relation to one another because no connective tissue directly articulates the bony surface.

Uniaxial Joint:-

Bony components are free to move in 1 plane around a single axis.

Hinge Joint -

Type of joint that represents door hinge

Ex - Elbow joint

- Ankle joint.

Interphalangeal joint

Pivot Joint -

Constructed so that one component is shaped like a ring and another is shaped so that it can rotate within the ring.

Ex:- Atlantoaxial joint.

Superior radioulnar joint

Bi-axial Joint :-

Bony components are free to move in 2 planes around two axis.

Condylar Joint -

Concave surface of the bony component is allowed to slide over the convex surface of another component.

Ex:- Knee joint

Temporomandibular joint

Saddle Joint -

Each joint surface is both convex in one plane & concave in the other, & these surfaces fit together like a rider on the saddle.

Ex:- Metacarpal phalangeal joint.

Tri-axial Joint :-

Bony components are free to move in 3 planes around 3 axis.

Ball and Socket Joint -

Ball like convex surface is fixed in the concave surface like socket.

Ex:- Shoulder joint

Hip joint

Plane Joint -

Plane joints have a variety of surface configurations & permits gliding movement.

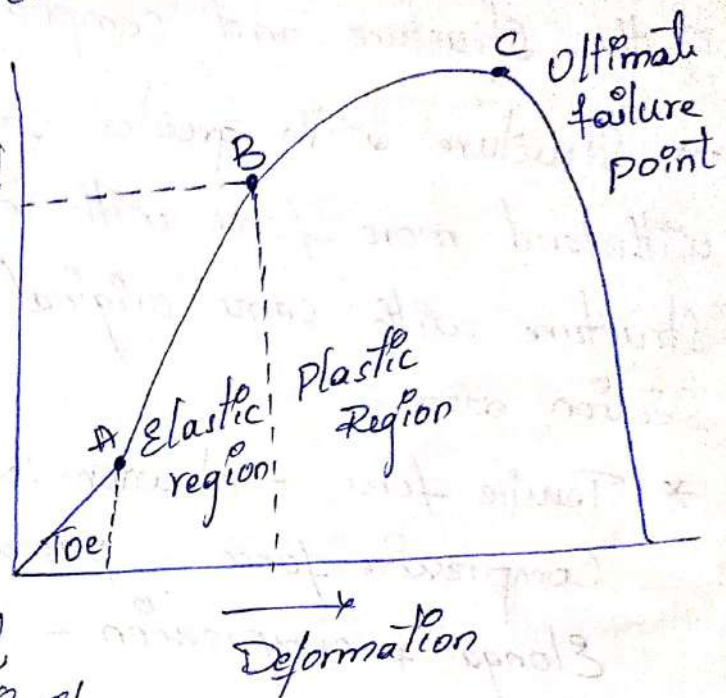
Ex:- Carpal & tarsal joints.

Load-deformation curve :-

- It is the result of plotting the applied force (load) against the deformation.
- It shows the elasticity, plasticity, ultimate strength and stiffness of the materials.

→ The portion of the curve b/w Point A & B is called Elastic region.

→ If the response to loading is confined to elastic region, then the deformation of the material is not permanent, the structure will regain its



original position after the force is removed.

→ The end point of elastic region is Yield point, at point B. The material will no longer immediately return to its original place but although it recovers in time.

→ The portion b/w point B & C is ultimate failure point Plastic region.

→ After the load is removed, the material will not be regain its original structure (deformation is permanent), but the recovery of the structure occurs in only synthesis and reorganization of new tissue components.

→ If loading continues through the plastic region, the material will continue to deform until it reaches the ultimate failure point 'c'.

→ Force value on the L-D curve depends on both the size of the structure and composition.

→ Structure with greater cross-section area can withstand more force with less deformation than the structure with same original length with less cross-section area.

* Tensile force - Newtons (N)

Compressive force - pascals (Pa)

Elongation & compression - Unit of length.

→ Greater cross-section area of tissue produces greater tensile strength (stiffness).

→ longer tissue produces less stiffness.